

Thermodynamics

Instructor

Research Assistant Buğra Kaplan

Chapter 2 – Problems

**Problem numbers correspond to
5th Edition**

2-26 Determine the energy required to accelerate an 800-kg car from rest to 100 km/h on a level road. *Answer: 309 kJ*

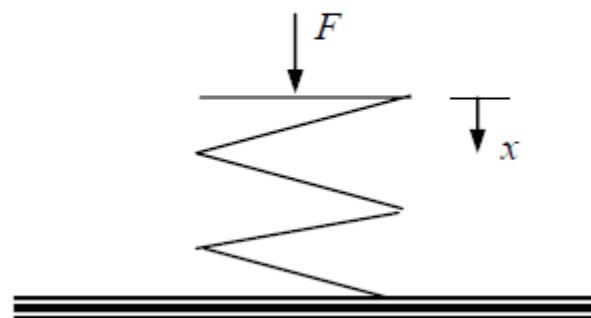
Analysis The work needed to accelerate a body the change in kinetic energy of the body,

$$W_a = \frac{1}{2} m(V_2^2 - V_1^2) = \frac{1}{2} (800 \text{ kg}) \left(\left(\frac{100,000 \text{ m}}{3600 \text{ s}} \right)^2 - 0 \right) \left(\frac{1 \text{ kJ}}{1000 \text{ kg} \cdot \text{m}^2/\text{s}^2} \right) = \mathbf{309 \text{ kJ}}$$

2-29 Determine the work required to deflect a linear spring with a spring constant of 70 kN/m by 20 cm from its rest position.

Analysis Since there is no preload, $F = kx$. Substituting this into the work expression gives

$$\begin{aligned} W &= \int_1^2 F ds = \int_1^2 kx dx = k \int_1^2 x dx = \frac{k}{2} (x_2^2 - x_1^2) \\ &= \frac{300 \text{ kN/m}}{2} [(0.03 \text{ m})^2 - 0^2] \\ &= 0.135 \text{ kN} \cdot \text{m} \\ &= (0.135 \text{ kN} \cdot \text{m}) \left(\frac{1 \text{ kJ}}{1 \text{ kN} \cdot \text{m}} \right) = \mathbf{0.135 \text{ kJ}} \end{aligned}$$



2–31 A ski lift has a one-way length of 1 km and a vertical rise of 200 m. The chairs are spaced 20 m apart, and each chair can seat three people. The lift is operating at a steady speed of 10 km/h. Neglecting friction and air drag and assuming that the average mass of each loaded chair is 250 kg, determine the power required to operate this ski lift. Also estimate the power required to accelerate this ski lift in 5 s to its operating speed when it is first turned on.

Assumptions 1 Air drag and friction are negligible. 2 The average mass of each loaded chair is 250 kg. 3 The mass of chairs is small relative to the mass of people, and thus the contribution of returning empty chairs to the motion is disregarded (this provides a safety factor).

Analysis The lift is 1000 m long and the chairs are spaced 20 m apart. Thus at any given time there are $1000/20 = 50$ chairs being lifted. Considering that the mass of each chair is 250 kg, the load of the lift at any given time is

$$\text{Load} = (50 \text{ chairs})(250 \text{ kg/chair}) = 12,500 \text{ kg}$$

Neglecting the work done on the system by the returning empty chairs, the work needed to raise this mass by 200 m is

$$W_g = mg(z_2 - z_1) = (12,500 \text{ kg})(9.81 \text{ m/s}^2)(200 \text{ m}) \left(\frac{1 \text{ kJ}}{1000 \text{ kg} \cdot \text{m}^2/\text{s}^2} \right) = 24,525 \text{ kJ}$$

At 10 km/h, it will take

$$\Delta t = \frac{\text{distance}}{\text{velocity}} = \frac{1 \text{ km}}{10 \text{ km/h}} = 0.1 \text{ h} = 360 \text{ s}$$

to do this work. Thus the power needed is

$$\dot{W}_g = \frac{W_g}{\Delta t} = \frac{24,525 \text{ kJ}}{360 \text{ s}} = \mathbf{68.1 \text{ kW}}$$

2-31 Continued

The velocity of the lift during steady operation, and the acceleration during start up are

$$V = (10 \text{ km/h}) \left(\frac{1 \text{ m/s}}{3.6 \text{ km/h}} \right) = 2.778 \text{ m/s}$$

$$a = \frac{\Delta V}{\Delta t} = \frac{2.778 \text{ m/s} - 0}{5 \text{ s}} = 0.556 \text{ m/s}^2$$

During acceleration, the power needed is

$$\dot{W}_a = \frac{1}{2} m (V_2^2 - V_1^2) / \Delta t = \frac{1}{2} (12,500 \text{ kg}) ((2.778 \text{ m/s})^2 - 0) \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) / (5 \text{ s}) = 9.6 \text{ kW}$$

Assuming the power applied is constant, the acceleration will also be constant and the vertical distance traveled during acceleration will be

$$h = \frac{1}{2} a t^2 \sin \alpha = \frac{1}{2} a t^2 \frac{200 \text{ m}}{1000 \text{ m}} = \frac{1}{2} (0.556 \text{ m/s}^2) (5 \text{ s})^2 (0.2) = 1.39 \text{ m}$$

and

$$\dot{W}_g = m g (z_2 - z_1) / \Delta t = (12,500 \text{ kg}) (9.81 \text{ m/s}^2) (1.39 \text{ m}) \left(\frac{1 \text{ kJ/kg}}{1000 \text{ kg} \cdot \text{m}^2/\text{s}^2} \right) / (5 \text{ s}) = 34.1 \text{ kW}$$

Thus,

$$\dot{W}_{\text{total}} = \dot{W}_a + \dot{W}_g = 9.6 + 34.1 = \mathbf{43.7 \text{ kW}}$$

2–32 Determine the power required for a 2000-kg car to climb a 100-m-long uphill road with a slope of 30° (from horizontal) in 10 s (a) at a constant velocity, (b) from rest to a final velocity of 30 m/s, and (c) from 35 m/s to a final velocity of 5 m/s. Disregard friction, air drag, and rolling resistance. **Answers: (a) 98.1 kW, (b) 188 kW, (c) –21.9 kW**

Assumptions Air drag, friction, and rolling resistance are negligible.

Analysis The total power required for each case is the sum of the rates of changes in potential and kinetic energies. That is,

$$\dot{W}_{\text{total}} = \dot{W}_a + \dot{W}_g$$

(a) $\dot{W}_a = 0$ since the velocity is constant. Also, the vertical rise is $h = (100 \text{ m})(\sin 30^\circ) = 50 \text{ m}$. Thus,

$$\dot{W}_g = mg(z_2 - z_1) / \Delta t = (2000 \text{ kg})(9.81 \text{ m/s}^2)(50 \text{ m}) \left(\frac{1 \text{ kJ}}{1000 \text{ kg} \cdot \text{m}^2/\text{s}^2} \right) / (10 \text{ s}) = 98.1 \text{ kW}$$

and $\dot{W}_{\text{total}} = \dot{W}_a + \dot{W}_g = 0 + 98.1 = \mathbf{98.1 \text{ kW}}$

(b) The power needed to accelerate is

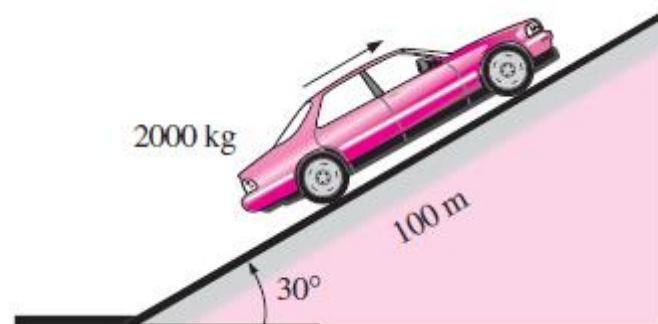
$$\dot{W}_a = \frac{1}{2} m(V_2^2 - V_1^2) / \Delta t = \frac{1}{2} (2000 \text{ kg}) [(30 \text{ m/s})^2 - 0] \left(\frac{1 \text{ kJ}}{1000 \text{ kg} \cdot \text{m}^2/\text{s}^2} \right) / (10 \text{ s}) = 90 \text{ kW}$$

and $\dot{W}_{\text{total}} = \dot{W}_a + \dot{W}_g = 90 + 98.1 = \mathbf{188.1 \text{ kW}}$

(c) The power needed to decelerate is

$$\dot{W}_a = \frac{1}{2} m(V_2^2 - V_1^2) / \Delta t = \frac{1}{2} (2000 \text{ kg}) [(5 \text{ m/s})^2 - (35 \text{ m/s})^2] \left(\frac{1 \text{ kJ}}{1000 \text{ kg} \cdot \text{m}^2/\text{s}^2} \right) / (10 \text{ s}) = -120 \text{ kW}$$

and $\dot{W}_{\text{total}} = \dot{W}_a + \dot{W}_g = -120 + 98.1 = \mathbf{-21.9 \text{ kW}}$ (braking power)



2-33 A damaged 1200-kg car is being towed by a truck. Neglecting the friction, air drag, and rolling resistance, determine the extra power required (a) for constant velocity on a level road, (b) for constant velocity of 50 km/h on a 30° (from horizontal) uphill road, and (c) to accelerate on a level road from stop to 90 km/h in 12 s. *Answers: (a) 0, (b) 81.7 kW, (c) 31.3 kW*

Assumptions Air drag, friction, and rolling resistance are negligible.

Analysis The total power required for each case is the sum of the rates of changes in potential and kinetic energies. That is,

$$\dot{W}_{\text{total}} = \dot{W}_a + \dot{W}_g$$

(a) Zero.

(b) $\dot{W}_a = 0$. Thus,

$$\begin{aligned}\dot{W}_{\text{total}} &= \dot{W}_g = mg(z_2 - z_1) / \Delta t = mg \frac{\Delta z}{\Delta t} = mgV_z = mgV \sin 30^\circ \\ &= (1200 \text{ kg})(9.81 \text{ m/s}^2) \left(\frac{50,000 \text{ m}}{3600 \text{ s}} \right) \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) (0.5) = \mathbf{81.7 \text{ kW}}\end{aligned}$$

(c) $\dot{W}_g = 0$. Thus,

$$\dot{W}_{\text{total}} = \dot{W}_a = \frac{1}{2} m(V_2^2 - V_1^2) / \Delta t = \frac{1}{2} (1200 \text{ kg}) \left(\left(\frac{90,000 \text{ m}}{3600 \text{ s}} \right)^2 - 0 \right) \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) / (12 \text{ s}) = \mathbf{31.3 \text{ kW}}$$



2-39 A classroom that normally contains 40 people is to be air-conditioned with window air-conditioning units of 5-kW cooling capacity. A person at rest may be assumed to dissipate heat at a rate of about 360 kJ/h. There are 10 light-bulbs in the room, each with a rating of 100 W. The rate of heat transfer to the classroom through the walls and the windows is estimated to be 15,000 kJ/h. If the room air is to be maintained at a constant temperature of 21°C, determine the number of window air-conditioning units required. *Answer:*

2 units

Assumptions There are no heat dissipating equipment (such as computers, TVs, or ranges) in the room.

Analysis The total cooling load of the room is determined from

$$\dot{Q}_{\text{cooling}} = \dot{Q}_{\text{lights}} + \dot{Q}_{\text{people}} + \dot{Q}_{\text{heat gain}}$$

where

$$\dot{Q}_{\text{lights}} = 10 \times 100 \text{ W} = 1 \text{ kW}$$

$$\dot{Q}_{\text{people}} = 40 \times 360 \text{ kJ/h} = 4 \text{ kW}$$

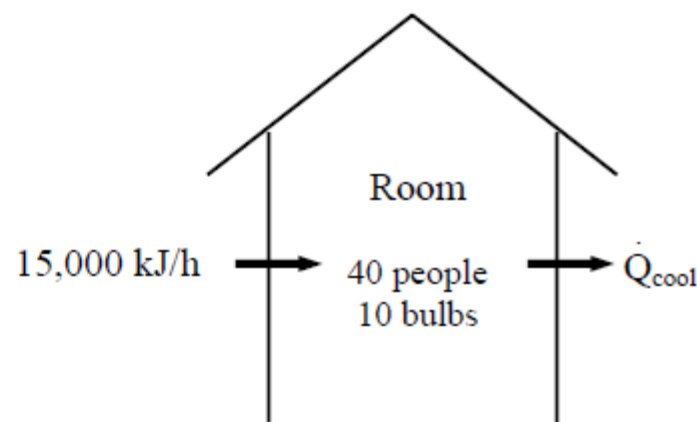
$$\dot{Q}_{\text{heat gain}} = 15,000 \text{ kJ/h} = 4.17 \text{ kW}$$

Substituting,

$$\dot{Q}_{\text{cooling}} = 1 + 4 + 4.17 = 9.17 \text{ kW}$$

Thus the number of air-conditioning units required is

$$\frac{9.17 \text{ kW}}{5 \text{ kW/unit}} = 1.83 \longrightarrow \mathbf{2 \text{ units}}$$



2-41 The lighting needs of a storage room are being met by 6 fluorescent light fixtures, each fixture containing four lamps rated at 60 W each. All the lamps are on during operating hours of the facility, which are 6 AM to 6 PM 365 days a year. The storage room is actually used for an average of 3 h a day. If the price of electricity is \$0.08/kWh, determine the amount of energy and money that will be saved as a result of installing motion sensors. Also, determine the simple payback period if the purchase price of the sensor is \$32 and it takes 1 hour to install it at a cost of \$40.

Assumptions The electrical energy consumed by the ballasts is negligible.

Analysis The plant operates 12 hours a day, and thus currently the lights are on for the entire 12 hour period. The motion sensors installed will keep the lights on for 3 hours, and off for the remaining 9 hours every day. This corresponds to a total of $9 \times 365 = 3285$ off hours per year. Disregarding the ballast factor, the annual energy and cost savings become

$$\begin{aligned}\text{Energy Savings} &= (\text{Number of lamps})(\text{Lamp wattage})(\text{Reduction of annual operating hours}) \\ &= (24 \text{ lamps})(60 \text{ W/lamp})(3285 \text{ hours/year}) \\ &= \mathbf{4730 \text{ kWh/year}}\end{aligned}$$

$$\begin{aligned}\text{Cost Savings} &= (\text{Energy Savings})(\text{Unit cost of energy}) \\ &= (4730 \text{ kWh/year})(\$0.08/\text{kWh}) \\ &= \mathbf{\$378/\text{year}}\end{aligned}$$

The implementation cost of this measure is the sum of the purchase price of the sensor plus the labor,

$$\text{Implementation Cost} = \text{Material} + \text{Labor} = \$32 + \$40 = \$72$$

This gives a simple payback period of

$$\text{Simple payback period} = \frac{\text{Implementation cost}}{\text{Annual cost savings}} = \frac{\$72}{\$378/\text{year}} = \mathbf{0.19 \text{ year}} \quad (2.3 \text{ months})$$

Therefore, the motion sensor will pay for itself in about 2 months.



2-42 A university campus has 200 classrooms and 400 faculty offices. The classrooms are equipped with 12 fluorescent tubes, each consuming 110 W, including the electricity used by the ballasts. The faculty offices, on average, have half as many tubes. The campus is open 240 days a year. The classrooms and faculty offices are not occupied an average of 4 h a day, but the lights are kept on. If the unit cost of electricity is \$0.082/kWh, determine how much the campus will save a year if the lights in the classrooms and faculty offices are turned off during unoccupied periods.

Analysis The total electric power consumed by the lights in the classrooms and faculty offices is

$$\dot{E}_{\text{lighting, classroom}} = (\text{Power consumed per lamp}) \times (\text{No. of lamps}) = (200 \times 12 \times 110 \text{ W}) = 264,000 = 264 \text{ kW}$$

$$\dot{E}_{\text{lighting, offices}} = (\text{Power consumed per lamp}) \times (\text{No. of lamps}) = (400 \times 6 \times 110 \text{ W}) = 264,000 = 264 \text{ kW}$$

$$\dot{E}_{\text{lighting, total}} = \dot{E}_{\text{lighting, classroom}} + \dot{E}_{\text{lighting, offices}} = 264 + 264 = 528 \text{ kW}$$

Noting that the campus is open 240 days a year, the total number of unoccupied work hours per year is

$$\text{Unoccupied hours} = (4 \text{ hours/day})(240 \text{ days/year}) = 960 \text{ h/yr}$$

Then the amount of electrical energy consumed per year during unoccupied work period and its cost are

$$\text{Energy savings} = (\dot{E}_{\text{lighting, total}})(\text{Unoccupied hours}) = (528 \text{ kW})(960 \text{ h/yr}) = 506,880 \text{ kWh}$$

$$\text{Cost savings} = (\text{Energy savings})(\text{Unit cost of energy}) = (506,880 \text{ kWh/yr})(\$0.082/\text{kWh}) = \mathbf{\$41,564/\text{yr}}$$

Discussion Note that simple conservation measures can result in significant energy and cost savings.

2-43 Consider a room that is initially at the outdoor temperature of 20°C. The room contains a 100-W lightbulb, a 110-W TV set, a 200-W refrigerator, and a 1000-W iron.

Assuming no heat transfer through the walls, determine the rate of increase of the energy content of the room when all of these electric devices are on.

Assumptions 1 The room is well sealed, and heat loss from the room is negligible. 2 All the appliances are kept on.

Analysis Taking the room as the system, the rate form of the energy balance can be written as

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{dE_{\text{system}} / dt}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \rightarrow dE_{\text{room}} / dt = \dot{E}_{in}$$

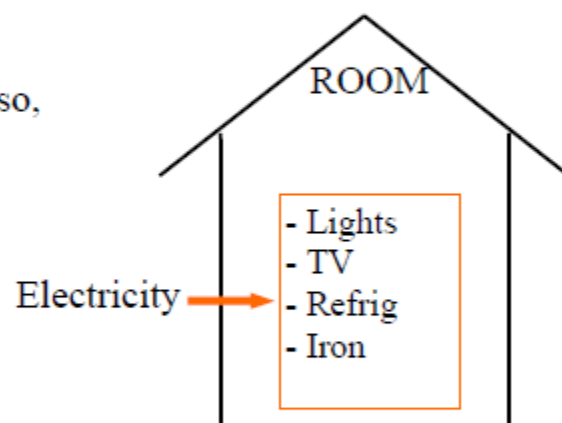
since no energy is leaving the room in any form, and thus $\dot{E}_{out} = 0$. Also,

$$\begin{aligned}\dot{E}_{in} &= \dot{E}_{\text{lights}} + \dot{E}_{\text{TV}} + \dot{E}_{\text{refrig}} + \dot{E}_{\text{iron}} \\ &= 100 + 110 + 200 + 1000 \text{ W} \\ &= 1410 \text{ W}\end{aligned}$$

Substituting, the rate of increase in the energy content of the room becomes

$$dE_{\text{room}} / dt = \dot{E}_{in} = \mathbf{1410 \text{ W}}$$

Discussion Note that some appliances such as refrigerators and irons operate intermittently, switching on and off as controlled by a thermostat. Therefore, the rate of energy transfer to the room, in general, will be less.



2-44 A fan is to accelerate quiescent air to a velocity of 10 m/s at a rate of 4 m³/s. Determine the minimum power that must be supplied to the fan. Take the density of air to be 1.18 kg/m³. *Answer: 236 W*

Assumptions The fan operates steadily.

Properties The density of air is given to be $\rho = 1.18 \text{ kg/m}^3$.

Analysis A fan transmits the mechanical energy of the shaft (shaft power) to mechanical energy of air (kinetic energy). For a control volume that encloses the fan, the energy balance can be written as

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{dE_{\text{system}} / dt}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{0 (steady)}}{=} 0 \quad \rightarrow \quad \dot{E}_{in} = \dot{E}_{out}$$

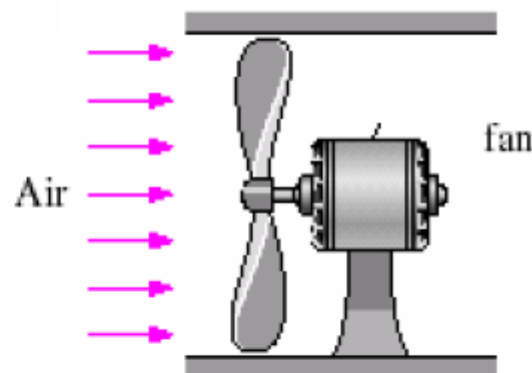
$$\dot{W}_{\text{sh}, in} = \dot{m}_{\text{air}} \text{ke}_{\text{out}} = \dot{m}_{\text{air}} \frac{V_{\text{out}}^2}{2}$$

where

$$\dot{m}_{\text{air}} = \rho \dot{V} = (1.18 \text{ kg/m}^3)(4 \text{ m}^3/\text{s}) = 4.72 \text{ kg/s}$$

Substituting, the minimum power input required is determined

$$\dot{W}_{\text{sh}, in} = \dot{m}_{\text{air}} \frac{V_{\text{out}}^2}{2} = (4.72 \text{ kg/s}) \frac{(10 \text{ m/s})^2}{2} \left(\frac{1 \text{ J/kg}}{1 \text{ m}^2/\text{s}^2} \right) = 236 \text{ J/s} = \mathbf{236 \text{ W}}$$



Discussion The conservation of energy principle requires the energy to be conserved as it is converted from one form to another, and it does not allow any energy to be created or destroyed during a process. In reality, the power required will be considerably higher because of the losses associated with the conversion of mechanical shaft energy to kinetic energy of air.

2-47



The driving force for fluid flow is the pressure difference, and a pump operates by raising the pressure of a fluid (by converting the mechanical shaft work to flow energy). A gasoline pump is measured to consume 5.2 kW of electric power when operating. If the pressure differential between the outlet and inlet of the pump is measured to be 5 kPa and the changes in velocity and elevation are negligible, determine the maximum possible volume flow rate of gasoline.

Assumptions 1 The gasoline pump operates steadily. 2 The changes in kinetic and potential energies across the pump are negligible.

Analysis For a control volume that encloses the pump-motor unit, the energy balance can be written as

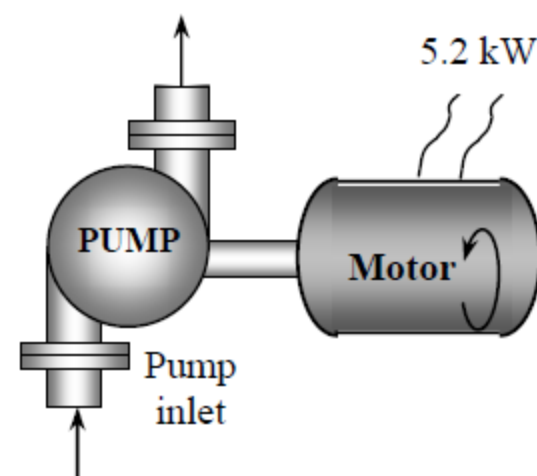
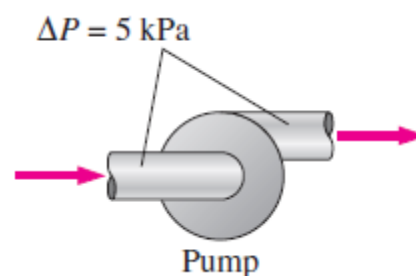
$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{dE_{\text{system}}/dt}_{\text{Rate of change in internal, kinetic, potential, etc. energies}}^{\phi=0 \text{ (steady)}} = 0 \rightarrow \dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{W}_{\text{in}} + \dot{m}(P\upsilon)_1 = \dot{m}(P\upsilon)_2 \rightarrow \dot{W}_{\text{in}} = \dot{m}(P_2 - P_1)\upsilon = \dot{V} \Delta P$$

since $\dot{m} = \dot{V}/\upsilon$ and the changes in kinetic and potential energies of gasoline are negligible, Solving for volume flow rate and substituting, the maximum flow rate is determined to be

$$\dot{V}_{\text{max}} = \frac{\dot{W}_{\text{in}}}{\Delta P} = \frac{5.2 \text{ kJ/s}}{5 \text{ kPa}} \left(\frac{1 \text{ kPa} \cdot \text{m}^3}{1 \text{ kJ}} \right) = 1.04 \text{ m}^3/\text{s}$$

Discussion The conservation of energy principle requires the energy to be conserved as it is converted from one form to another, and it does not allow any energy to be created or destroyed during a process. In reality, the volume flow rate will be less because of the losses associated with the conversion of electrical-to-mechanical shaft and mechanical shaft-to-flow energy.



2-50 An escalator in a shopping center is designed to move 30 people, 75 kg each, at a constant speed of 0.8 m/s at 45° slope. Determine the minimum power input needed to drive this escalator. What would your answer be if the escalator velocity were to be doubled?

Assumptions 1 Air drag and friction are negligible. 2 The average mass of each person is 75 kg. 3 The escalator operates steadily, with no acceleration or breaking. 4 The mass of escalator itself is negligible.

Analysis At design conditions, the total mass moved by the escalator at any given time is

$$\text{Mass} = (30 \text{ persons})(75 \text{ kg/person}) = 2250 \text{ kg}$$

The vertical component of escalator velocity is

$$V_{\text{vert}} = V \sin 45^\circ = (0.8 \text{ m/s}) \sin 45^\circ$$

Under stated assumptions, the power supplied is used to increase the potential energy of people. Taking the people on elevator as the closed system, the energy balance in the rate form can be written as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{dE_{\text{system}}/dt}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} = 0 \rightarrow \dot{E}_{\text{in}} = dE_{\text{sys}}/dt \cong \frac{\Delta E_{\text{sys}}}{\Delta t}$$

$$\dot{W}_{\text{in}} = \frac{\Delta PE}{\Delta t} = \frac{mg\Delta z}{\Delta t} = mgV_{\text{vert}}$$

That is, under stated assumptions, the power input to the escalator must be equal to the rate of increase of the potential energy of people. Substituting, the required power input becomes

$$\dot{W}_{\text{in}} = mgV_{\text{vert}} = (2250 \text{ kg})(9.81 \text{ m/s}^2)(0.8 \text{ m/s}) \sin 45^\circ \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = 12.5 \text{ kJ/s} = \mathbf{12.5 \text{ kW}}$$

When the escalator velocity is doubled to $V = 1.6 \text{ m/s}$, the power needed to drive the escalator becomes

$$\dot{W}_{\text{in}} = mgV_{\text{vert}} = (2250 \text{ kg})(9.81 \text{ m/s}^2)(1.6 \text{ m/s}) \sin 45^\circ \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = 25.0 \text{ kJ/s} = \mathbf{25.0 \text{ kW}}$$

Discussion Note that the power needed to drive an escalator is proportional to the escalator velocity.

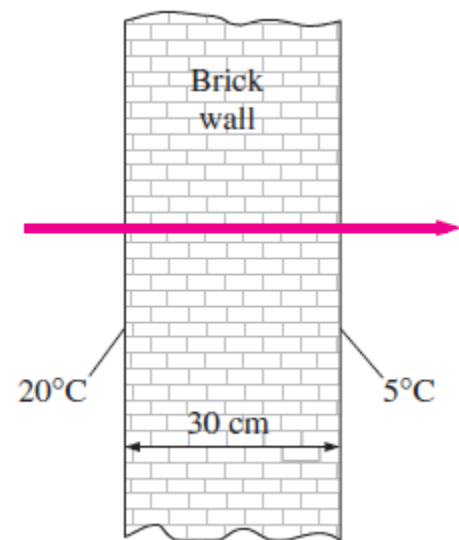
2–95 The inner and outer surfaces of a 5-m $\times$ 6-m brick wall of thickness 30 cm and thermal conductivity 0.69 W/m $\cdot$ $^{\circ}$ C are maintained at temperatures of 20 $^{\circ}$ C and 5 $^{\circ}$ C, respectively. Determine the rate of heat transfer through the wall, in W.

Assumptions 1 Steady operating conditions exist since the surface temperatures of the wall remain constant at the specified values. 2 Thermal properties of the wall are constant.

Properties The thermal conductivity of the wall is given to be $k = 0.69$ W/m $\cdot$ $^{\circ}$ C.

Analysis Under steady conditions, the rate of heat transfer through the wall is

$$\dot{Q}_{\text{cond}} = kA \frac{\Delta T}{L} = (0.69 \text{ W/m} \cdot ^{\circ}\text{C})(5 \times 6 \text{ m}^2) \frac{(20 - 5)^{\circ}\text{C}}{0.3 \text{ m}} = \mathbf{1035 \text{ W}}$$



2-96 The inner and outer surfaces of a 0.5-cm-thick 2-m $\times$ 2-m window glass in winter are 10°C and 3°C, respectively. If the thermal conductivity of the glass is 0.78 W/m $\cdot$ °C, determine the amount of heat loss, in kJ, through the glass over a period of 5 h. What would your answer be if the glass were 1-cm thick?

Assumptions 1 Steady operating conditions exist since the surface temperatures of the glass remain constant at the specified values. 2 Thermal properties of the glass are constant.

Properties The thermal conductivity of the glass is given to be $k = 0.78$ W/m $\cdot$ °C.

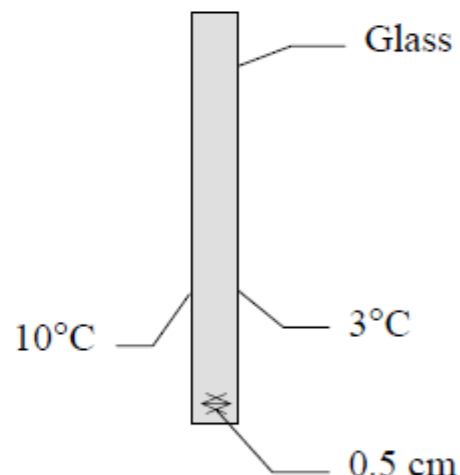
Analysis Under steady conditions, the rate of heat transfer through the glass by conduction is

$$\dot{Q}_{\text{cond}} = kA \frac{\Delta T}{L} = (0.78 \text{ W/m} \cdot ^\circ\text{C})(2 \times 2 \text{ m}^2) \frac{(10 - 3)^\circ\text{C}}{0.005 \text{ m}} = 4368 \text{ W}$$

Then the amount of heat transferred over a period of 5 h becomes

$$Q = \dot{Q}_{\text{cond}} \Delta t = (4.368 \text{ kJ/s})(5 \times 3600 \text{ s}) = \mathbf{78,600 \text{ kJ}}$$

If the thickness of the glass is doubled to 1 cm, then the amount of heat transferred will go down by half to **39,300 kJ**.



2-98 An aluminum pan whose thermal conductivity is $237 \text{ W/m} \cdot ^\circ\text{C}$ has a flat bottom whose diameter is 20 cm and thickness 0.4 cm. Heat is transferred steadily to boiling water in the pan through its bottom at a rate of 500 W. If the inner surface of the bottom of the pan is 105°C , determine the temperature of the outer surface of the bottom of the pan.

Assumptions 1 Steady operating conditions exist since the surface temperatures of the pan remain constant at the specified values. 2 Thermal properties of the aluminum pan are constant.

Properties The thermal conductivity of the aluminum is given to be $k = 237 \text{ W/m} \cdot ^\circ\text{C}$.

Analysis The heat transfer surface area is

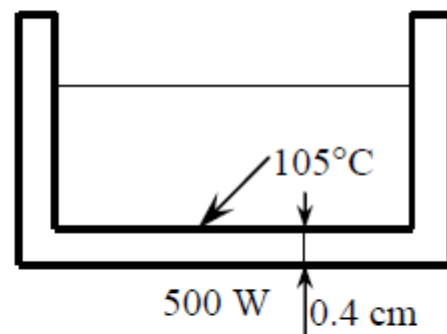
$$A = \pi r^2 = \pi(0.1 \text{ m})^2 = 0.0314 \text{ m}^2$$

Under steady conditions, the rate of heat transfer through the bottom of the pan by conduction is

$$\dot{Q} = kA \frac{\Delta T}{L} = kA \frac{T_2 - T_1}{L}$$

Substituting, $500 \text{ W} = (237 \text{ W/m} \cdot ^\circ\text{C})(0.0314 \text{ m}^2) \frac{T_2 - 105^\circ\text{C}}{0.004 \text{ m}}$

which gives $T_2 = 105.3^\circ\text{C}$



2-99 For heat transfer purposes, a standing man can be modeled as a 30-cm diameter, 170-cm long vertical cylinder with both the top and bottom surfaces insulated and with the side surface at an average temperature of 34°C. For a convection heat transfer coefficient of 15 W/m² · °C, determine the rate of heat loss from this man by convection in an environment at 20°C. *Answer: 336 W*

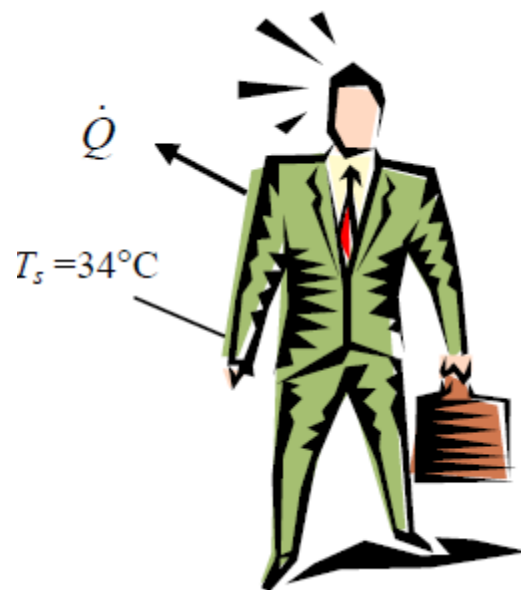
Assumptions 1 Steady operating conditions exist. 2 Heat transfer by radiation is not considered. 3 The environment is at a uniform temperature.

Analysis The heat transfer surface area of the person is

$$A = \pi DL = \pi(0.3 \text{ m})(1.70 \text{ m}) = 1.60 \text{ m}^2$$

Under steady conditions, the rate of heat transfer by convection is

$$\dot{Q}_{\text{conv}} = hA\Delta T = (15 \text{ W/m}^2 \cdot ^\circ\text{C})(1.60 \text{ m}^2)(34 - 20)^\circ\text{C} = \mathbf{336 \text{ W}}$$



2-100 A 5-cm-diameter spherical ball whose surface is maintained at a temperature of 70°C is suspended in the middle of a room at 20°C. If the convection heat transfer coefficient is 15 W/m² · °C and the emissivity of the surface is 0.8, determine the total rate of heat transfer from the ball.

Assumptions 1 Steady operating conditions exist since the ball surface and the surrounding air and surfaces remain at constant temperatures. 2 The thermal properties of the ball and the convection heat transfer coefficient are constant and uniform.

Properties The emissivity of the ball surface is given to be $\varepsilon = 0.8$.

Analysis The heat transfer surface area is

$$A = \pi D^2 = 3.14 \times (0.05 \text{ m})^2 = 0.007854 \text{ m}^2$$

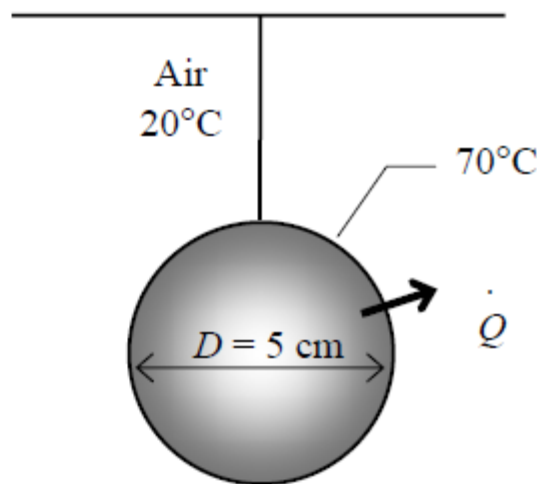
Under steady conditions, the rates of convection and radiation heat transfer are

$$\dot{Q}_{\text{conv}} = hA\Delta T = (15 \text{ W/m}^2 \cdot ^\circ\text{C})(0.007854 \text{ m}^2)(70 - 20)^\circ\text{C} = 5.89 \text{ W}$$

$$\dot{Q}_{\text{rad}} = \varepsilon \sigma A (T_s^4 - T_o^4) = 0.8(0.007854 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(343 \text{ K})^4 - (293 \text{ K})^4] = 2.31 \text{ W}$$

Therefore,

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = 5.89 + 2.31 = \mathbf{8.20 \text{ W}}$$

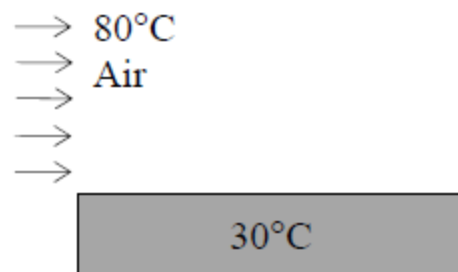


2-102 Hot air at 80°C is blown over a $2\text{-m} \times 4\text{-m}$ flat surface at 30°C . If the convection heat transfer coefficient is $55\text{ W/m}^2 \cdot ^{\circ}\text{C}$, determine the rate of heat transfer from the air to the plate, in kW.

Assumptions **1** Steady operating conditions exist. **2** Heat transfer by radiation is not considered. **3** The convection heat transfer coefficient is constant and uniform over the surface.

Analysis Under steady conditions, the rate of heat transfer by convection is

$$\begin{aligned}\dot{Q}_{\text{conv}} &= hA\Delta T \\ &= (55\text{ W/m}^2 \cdot ^{\circ}\text{C})(2 \times 4\text{ m}^2)(80 - 30)^{\circ}\text{C} \\ &= \mathbf{22,000\text{ W} = 22\text{ kW}}\end{aligned}$$



2-103 A 1000-W iron is left on the ironing board with its base exposed to the air at 20°C. The convection heat transfer coefficient between the base surface and the surrounding air is 35 W/m² · °C. If the base has an emissivity of 0.6 and a surface area of 0.02 m², determine the temperature of the base of the iron.

Assumptions 1 Steady operating conditions exist. 2 The thermal properties of the iron base and the convection heat transfer coefficient are constant and uniform. 3 The temperature of the surrounding surfaces is the same as the temperature of the surrounding air.

Properties The emissivity of the base surface is given to be $\varepsilon = 0.6$.

Analysis At steady conditions, the 1000 W of energy supplied to the iron will be dissipated to the surroundings by convection and radiation heat transfer. Therefore,

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = 1000 \text{ W}$$

where $\dot{Q}_{\text{conv}} = hA\Delta T = (35 \text{ W/m}^2 \cdot \text{K})(0.02 \text{ m}^2)(T_s - 293 \text{ K}) = 0.7(T_s - 293 \text{ K}) \text{ W}$

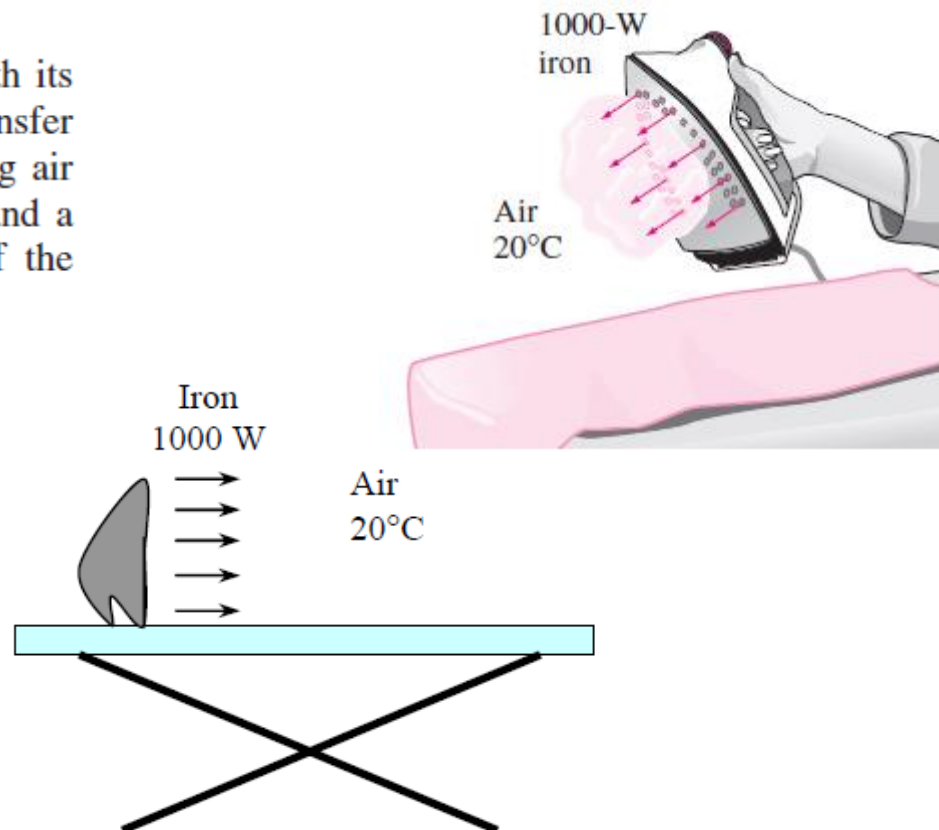
and

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon\sigma A(T_s^4 - T_o^4) = 0.6(0.02 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[T_s^4 - (293 \text{ K})^4] \\ &= 0.06804 \times 10^{-8}[T_s^4 - (293 \text{ K})^4] \text{ W} \end{aligned}$$

Substituting, $1000 \text{ W} = 0.7(T_s - 293 \text{ K}) + 0.06804 \times 10^{-8}[T_s^4 - (293 \text{ K})^4]$

Solving by trial and error gives $T_s = \mathbf{947 \text{ K} = 674^\circ\text{C}}$

Discussion We note that the iron will dissipate all the energy it receives by convection and radiation when its surface temperature reaches 947 K.



2-104 A thin metal plate is insulated on the back and exposed to solar radiation on the front surface. The exposed surface of the plate has an absorptivity of 0.6 for solar radiation. If solar radiation is incident on the plate at a rate of 700 W/m^2 and the surrounding air temperature is 25°C , determine the surface temperature of the plate when the heat loss by convection equals the solar energy absorbed by the plate. Assume the convection heat transfer coefficient to be $50 \text{ W/m}^2 \cdot ^\circ\text{C}$, and disregard heat loss by radiation.

Assumptions 1 Steady operating conditions exist. 2 Heat transfer through the insulated side of the plate is negligible. 3 The heat transfer coefficient is constant and uniform over the plate. 4 Heat loss by radiation is negligible.

Properties The solar absorptivity of the plate is given to be $\alpha = 0.6$.

Analysis When the heat loss from the plate by convection equals the solar radiation absorbed, the surface temperature of the plate can be determined from

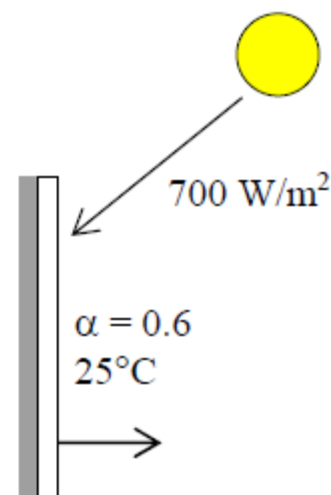
$$\dot{Q}_{\text{solar absorbed}} = \dot{Q}_{\text{conv}}$$

$$\alpha \dot{Q}_{\text{solar}} = hA(T_s - T_o)$$

$$0.6 \times A \times 700 \text{ W/m}^2 = (50 \text{ W/m}^2 \cdot ^\circ\text{C})A(T_s - 25)$$

Canceling the surface area A and solving for T_s gives

$$T_s = \mathbf{33.4^\circ\text{C}}$$



2-106 A 5-cm-external-diameter, 10-m-long hot-water pipe at 80°C is losing heat to the surrounding air at 5°C by natural convection with a heat transfer coefficient of 25 W/m² · °C. Determine the rate of heat loss from the pipe by natural convection, in kW.

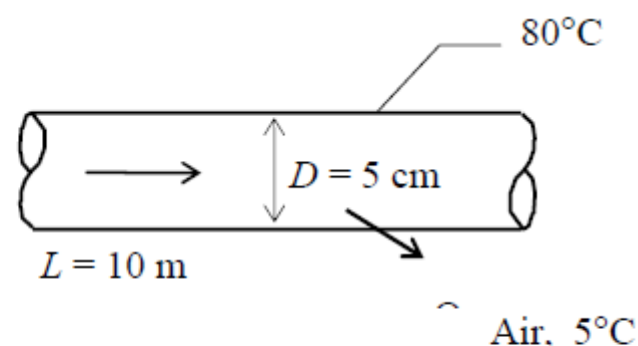
Assumptions 1 Steady operating conditions exist. 2 Heat transfer by radiation is not considered. 3 The convection heat transfer coefficient is constant and uniform over the surface.

Analysis The heat transfer surface area is

$$A = (\pi D)L = 3.14 \times (0.05 \text{ m})(10 \text{ m}) = 1.571 \text{ m}^2$$

Under steady conditions, the rate of heat transfer by convection is

$$\dot{Q}_{\text{conv}} = hA\Delta T = (25 \text{ W/m}^2 \cdot ^\circ\text{C})(1.571 \text{ m}^2)(80 - 5)^\circ\text{C} = \mathbf{2945 \text{ W} = 2.95 \text{ kW}}$$



2-107 The outer surface of a spacecraft in space has an emissivity of 0.8 and an absorptivity of 0.3 for solar radiation. If solar radiation is incident on the spacecraft at a rate of 1000 W/m^2 , determine the surface temperature of the spacecraft when the radiation emitted equals the solar energy absorbed.

Assumptions 1 Steady operating conditions exist since the surface temperatures of the wall remain constant at the specified values. 2 Thermal properties of the spacecraft are constant.

Properties The outer surface of a spacecraft has an emissivity of 0.8 and an absorptivity of 0.3.

Analysis When the heat loss from the outer surface of the spacecraft by radiation equals the solar radiation absorbed, the surface temperature can be determined from

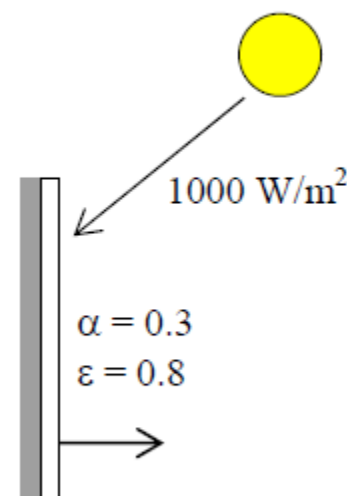
$$\dot{Q}_{\text{solar absorbed}} = \dot{Q}_{\text{rad}}$$

$$\alpha \dot{Q}_{\text{solar}} = \varepsilon \sigma A (T_s^4 - T_{\text{space}}^4)$$

$$0.3 \times A \times (1000 \text{ W/m}^2) = 0.8 \times A \times (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [T_s^4 - (0 \text{ K})^4]$$

Canceling the surface area A and solving for T_s gives

$$T_s = \mathbf{285 \text{ K}}$$



2-109 A hollow spherical iron container whose outer diameter is 20 cm and thickness is 0.4 cm is filled with iced water at 0°C. If the outer surface temperature is 5°C, determine the approximate rate of heat loss from the sphere, and the rate at which ice melts in the container.

Assumptions 1 Steady operating conditions exist since the surface temperatures of the wall remain constant at the specified values. 2 Heat transfer through the shell is one-dimensional. 3 Thermal properties of the iron shell are constant. 4 The inner surface of the shell is at the same temperature as the iced water, 0°C.

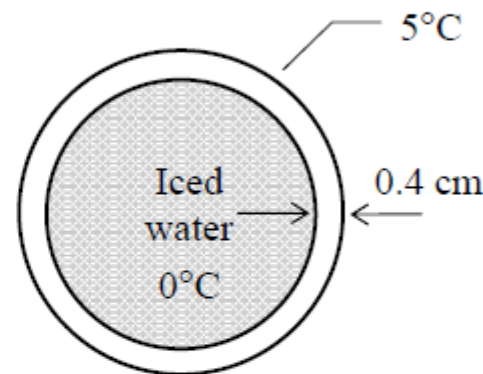
Properties The thermal conductivity of iron is $k = 80.2 \text{ W/m}\cdot^\circ\text{C}$ (Table 2-3). The heat of fusion of water is at 1 atm is 333.7 kJ/kg.

Analysis This spherical shell can be approximated as a plate of thickness 0.4 cm and surface area

$$A = \pi D^2 = 3.14 \times (0.2 \text{ m})^2 = 0.126 \text{ m}^2$$

Then the rate of heat transfer through the shell by conduction is

$$\dot{Q}_{\text{cond}} = kA \frac{\Delta T}{L} = (80.2 \text{ W/m}\cdot^\circ\text{C})(0.126 \text{ m}^2) \frac{(5 - 0)^\circ\text{C}}{0.004 \text{ m}} = \mathbf{12,632 \text{ W}}$$



Considering that it takes 333.7 kJ of energy to melt 1 kg of ice at 0°C, the rate at which ice melts in the container can be determined from

$$\dot{m}_{\text{ice}} = \frac{\dot{Q}}{h_{\text{if}}} = \frac{12.632 \text{ kJ/s}}{333.7 \text{ kJ/kg}} = \mathbf{0.038 \text{ kg/s}}$$

Discussion We should point out that this result is slightly in error for approximating a curved wall as a plain wall. The error in this case is very small because of the large diameter to thickness ratio. For better accuracy, we could use the inner surface area ($D = 19.2 \text{ cm}$) or the mean surface area ($D = 19.6 \text{ cm}$) in the calculations.

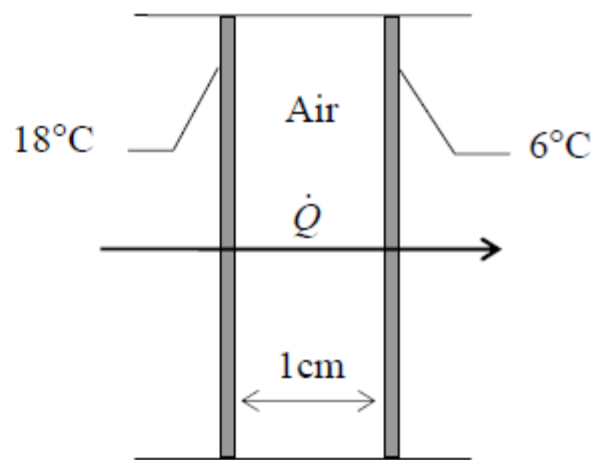
2-110 The inner and outer glasses of a $2\text{-m} \times 2\text{-m}$ double pane window are at 18°C and 6°C , respectively. If the 1-cm space between the two glasses is filled with still air, determine the rate of heat transfer through the window, in kW.

Assumptions 1 Steady operating conditions exist since the surface temperatures of the glass remain constant at the specified values. 2 Heat transfer through the window is one-dimensional. 3 Thermal properties of the air are constant. 4 The air trapped between the two glasses is still, and thus heat transfer is by conduction only.

Properties The thermal conductivity of air at room temperature is $k = 0.026 \text{ W/m}\cdot^\circ\text{C}$ (Table 2-3).

Analysis Under steady conditions, the rate of heat transfer through the window by conduction is

$$\begin{aligned}\dot{Q}_{\text{cond}} &= kA \frac{\Delta T}{L} = (0.026 \text{ W/m}\cdot^\circ\text{C})(2 \times 2 \text{ m}^2) \frac{(18 - 6)^\circ\text{C}}{0.01 \text{ m}} \\ &= \mathbf{125 \text{ W} = 0.125 \text{ kW}}\end{aligned}$$



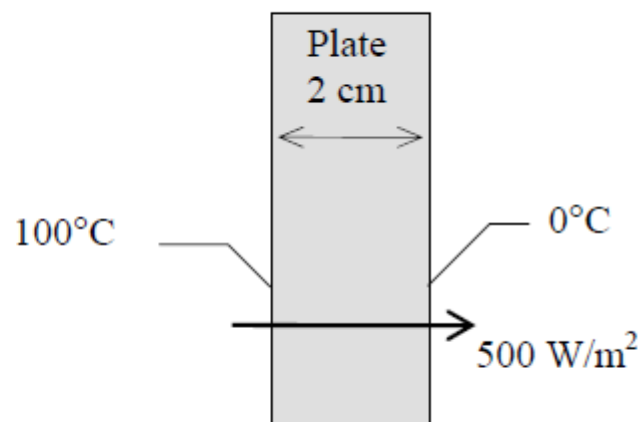
2-111 Two surfaces of a 2-cm-thick plate are maintained at 0°C and 100°C, respectively. If it is determined that heat is transferred through the plate at a rate of 500 W/m², determine its thermal conductivity.

Assumptions 1 Steady operating conditions exist since the surface temperatures of the plate remain constant at the specified values. 2 Heat transfer through the plate is one-dimensional. 3 Thermal properties of the plate are constant.

Analysis The thermal conductivity is determined directly from the steady one-dimensional heat conduction relation to be

$$\dot{Q} = kA \frac{T_1 - T_2}{L}$$

$$k = \frac{(\dot{Q}/A)L}{T_1 - T_2} = \frac{(500 \text{ W/m}^2)(0.02 \text{ m})}{(100 - 0)^\circ\text{C}} = \mathbf{0.1 \text{ W/m}\cdot^\circ\text{C}}$$



Chapter 2

End of the Problems